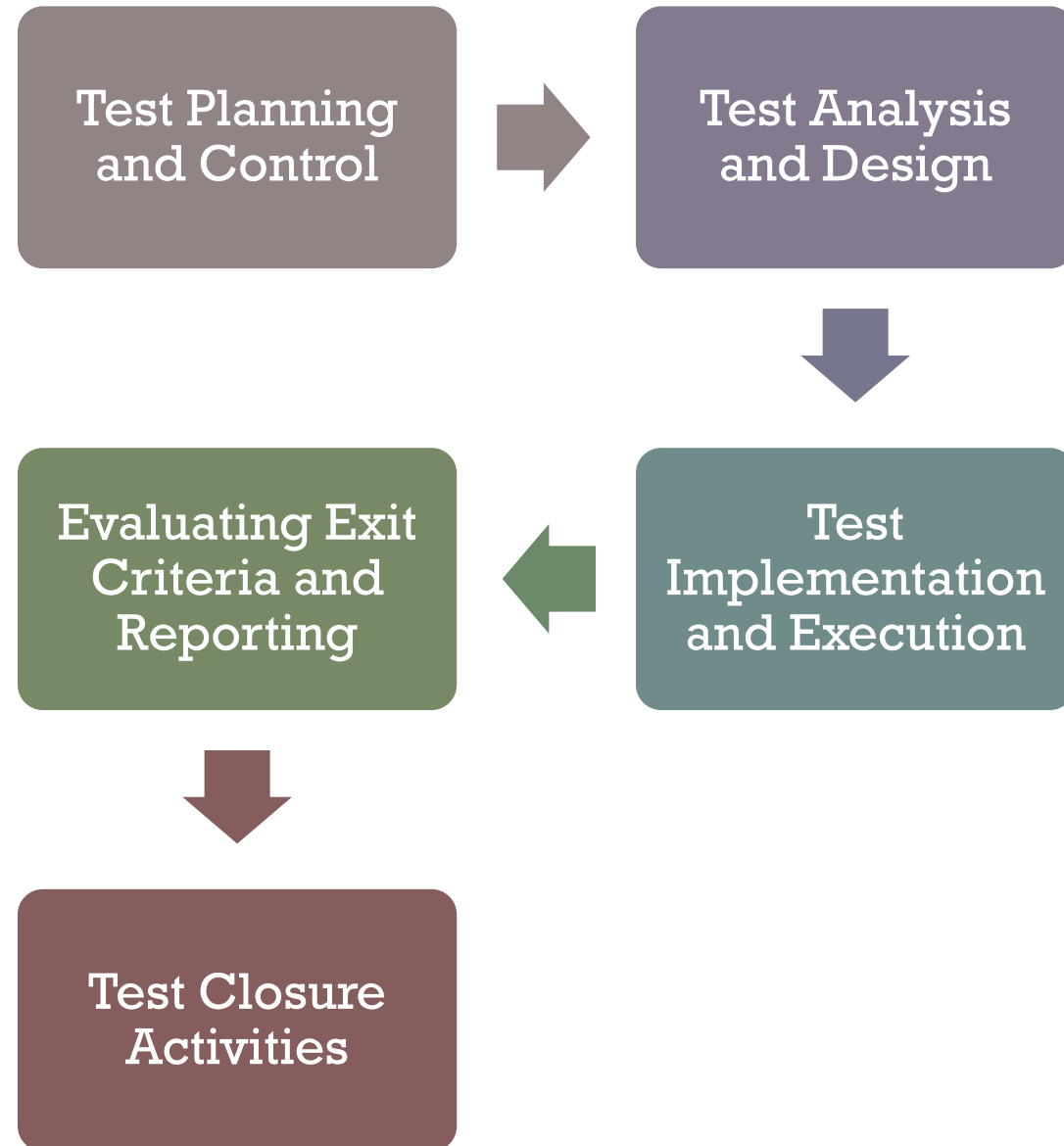


TESTING AND INTEGRATION REQUIREMENT TRACING

DEFINITION OF TESTING

- S/W testing is process, to evaluate the functionality of a software application with an intent to find if the developed S/W meet the requirements or not
- It is defined as an execution of a work product with intent to find defects
- Testing is done to protect the common users from any failure of system during usage.
- The process can include:
 - Verifying S/W completeness in regards to functional/business requirements
 - Identifying technical bugs/errors and ensuring the software is error free
 - Assessing usability, performance, security, localization, compatibility and installation



FUNDAMENTAL TEST PROCESS

THE PSYCHOLOGY OF TESTING

- Comparison of the mindset of the tester and developer
- The balance between self-testing and independent Testing
- There should be clear and courteous communication and feedback on defects between tester and developer



TESTING APPROACHES

1. Bing bang approach

- It involves testing S/W after completion of development
- It also called 'Final Testing' or 'System Testing'
- It is done before releasing S/W
- Bing bang approach has main trust on black box testing of S/W to ensure that the requirements as defined and documented in requirement and design specifications are met successfully.
- In Bing bang approach, S/W is tested before delivery
- It not possible to detect all faults due to time constraints

TESTING APPROACHES CONT.

2. Total Quality Management Approach

- TQM can be defined as management technique for improving processes, products and services
- It focuses on the entire business and NOT just on particular process
- Defect removal costs are approximately 10 times more after coding
- This is a cost associated with fixing the problems belonging to requirements, design, coding etc.

REQUIREMENT TRACEABILITY MATRIX(RTM)

- RTM is used to trace the requirements to the tests they are needed to verify whether the requirements are fulfilled
- The purpose of the RTM is to ensure that all requirements defined for a system are tested.
- The RTM is usually developed in concurrency with the initial list of requirements.
- Types of RTM:
 - Forward Traceability(Mapping requirements to the test cases)
 - Backward or Reverse Traceability(Mapping test cases to requirements)

Req No	Req Desc	Testcase ID	Status
123	Login to the application	TC01,TC02,TC03	TC01-Pass TC02-Pass
345	Ticket Creation	TC04,TC05,TC06,TC07,TC08,TC09 TC010	TC04-Pass TC05-Pass TC06-Pass TC06-Fail TC07-No Run
456	Search Ticket	TC011,TC012,TC013,TC014	TC011-Pass TC012-Fail TC013-Pass TC014-No Run

REQUIREMENT TRACEABILITY MATRIX(RTM) CONT.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
2																
3		Sno	Req ID	Req Desc	TC ID	TC Desc	Test Design	Test Designer	UAT Test Req?	Test Execution			Defects?	Defect ID	Defect Status	Req Coverage Status
4										Test Env	UAT Env	Prod Env				
5		1	Req01	Login to the Application	TC01	Login with Invalid Username and valid password	Completed	XYZ	No	Passed	No Run	No Run	None	None	N/A	Partial
6		2			TC02	Login with Valid Username and invalid password	Completed	YZA	No	Passed	No Run	No Run	None	None	N/A	Partial
7		3			TC03	Login with valid credentials	Completed	XYZ	Yes	Passed	Passed	No Run	Yes	DFCT001	Test OK	Partial
8																

ADVANTAGES OF RTM

- Entire S/W development can be tracked completely through RTM
- Any test case failure can be tracked through requirements, designs, coding etc.
- Any changes in requirements can be affected through entire work product upto test cases
- Application becomes maintainable
- Its also helps to understand if any redundancy has been created in the application
- It may answer questions about what is being developed

PROBLEMS WITH RTM

- Number of requirements are huge
- Connectivity with different relationship
- Requirements changes frequently and so do designs, code and test cases
- A customer may not find value in it and pay for it
- Incremental and iterative developments are the major challenges for maintaining traceability

ESSENTIALS OF SOFTWARE TESTING

- Test Strategy
- Strengths
- Weaknesses
- Testing Plan
- Test cases
- Opportunities
- Test data
- Threats
- Test Environment

WORKBENCH

A workbench concept is a method of scheduling that how a particular action has to be executed. Actions contain in workbench are:

- Input
- Execute
- Check
- Production Output
- Rework

IMPORTANT FEATURES OF TESTING PROCESS

- Easy to operate
- Stability
- Observability
- Easy to understand
- Decomposability

MISCONCEPTIONS ABOUT TESTING

- Anyone can test
- Testing can happen at the end
- Testing can find all bugs
- Defects found by customers are blamed on Tester
- Test automation will replace manual testers.

PRINCIPLES OF SOFTWARE TESTING

- Testing shows presence of defects
- Exhaustive testing is not possible
- Early Testing
- Defect Clustering
- Pesticide Paradox
- Testing is context dependent
- Absence of errors fallacy

SALIENT FEATURES OF GOOD TESTING

- Capturing user requirements
- Be Sceptical
- Don't compromise on quality
- Design objectives
- Ensure end user satisfaction
- Think from the users perspective
- Never Promise 100% coverage
- Start early
- Be open Suggestions

TEST POLICY

- It is a document described at the organization level
- It is determined by the senior management of the organization
- The test policy is very high-level document and at the top of the test documentation structure
- Organizations may prefer to publish their test policy in a sentence as well as a separate document
- Policy can be used in both development and maintenance project

TEST POLICY SHALL DESCRIBE THE FOLLOWINGS

- Clear answer to the questions
- It is determined by the senior management of the organisation.
- The definition of the testing process used by the organization to increase the quality of the software developed
- How the organization will measure the effectiveness and efficiency of the test while achieving goals.
- How the organization will improve its test processes

TEST STRATEGY

- It is prepared at the program level and includes general test strategy, management principles, processes and approaches
- It is also a high level document and is usually written by the test manager
- It is generally prepared by the large scale projects in small scale projects, test strategies and test approach may be included in the test plan

TEST STRATEGY INCLUDES:



Objectives/
Scope of
testing



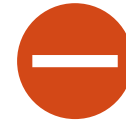
In-scope and
Out-scope



Test Levels



Test
types(Function
/non-
functional)



Entry/Exit



Risk



Test
environment



Retesting &
Regression
approach

TEST PLANNING

- All test strategies specific to the project
- Test estimations and test schedule
- Test organization/roles/responsibilities
- Test deliverables
- Test reporting principles

TESTING PROCESS

- Testing is a process rather than a single activity
- Test processes are a vital part of SDLC
- From planning to execution, each stage of the process is systematically planned
- This can be combined and evaluated to get expected results and outcomes
- The quality and effectiveness of the S/W testing is primarily determined by the quality of the test process

TEST TEAM EFFICIENCY

- It is a very important aspect for development team and management
- If team is very efficient, less iterations of testing are required
- Every test manager must be aware of the efficiency of his team
- Test team efficiency is depend on organizational culture

MUTATION TESTING

- It is used to check capability of test program and test cases to find defects
- The mutations introduced to source code are designed to imitate common programming errors
- A good unit test suite typically detects the program mutations and fails automatically
- The changes in mutant program are kept extremely small, so it does not affect the overall objective of the program

CHALLENGES IN TESTING

Complete testing is impossible

Setting the right process

Lack of communication

Lack of recourses

Test coverage

TEST TEAM APPROACH

Proactive communication

Role and responsibility of Team members

Program Manager

QA Lead

Test analyst

Test Engineer

PROCESS PROBLEMS FACED BY TESTING

- Unstable Environment
- Tools being force-fed
- Lack of feedback loop
- Preconceived notions
- No documentation

COST ASPECT OF TESTING



Prevention
Cost



Appraisal
Cost



Internal
failure cost



External
Failure Cost

CONTENTS OF TEST POLICY DOCUMENT

Definition of Testing

Description of Test process

Test evaluation

Quality Level to be achieved

Approach to Test process improvement

ERRORS

Errors can be present in software due to following reasons

- Programming Error
- Unclear requirements
- Changing requirements
- Software complexity
- Time Pressures
- Poorly Documented Codes

DEFECTS

- Defects can be simply defined as a variance between expected and actual
- Defects is an error found after the application goes into production
- Defects can be divided as follows:
 - Wrong
 - Missing
 - Extra

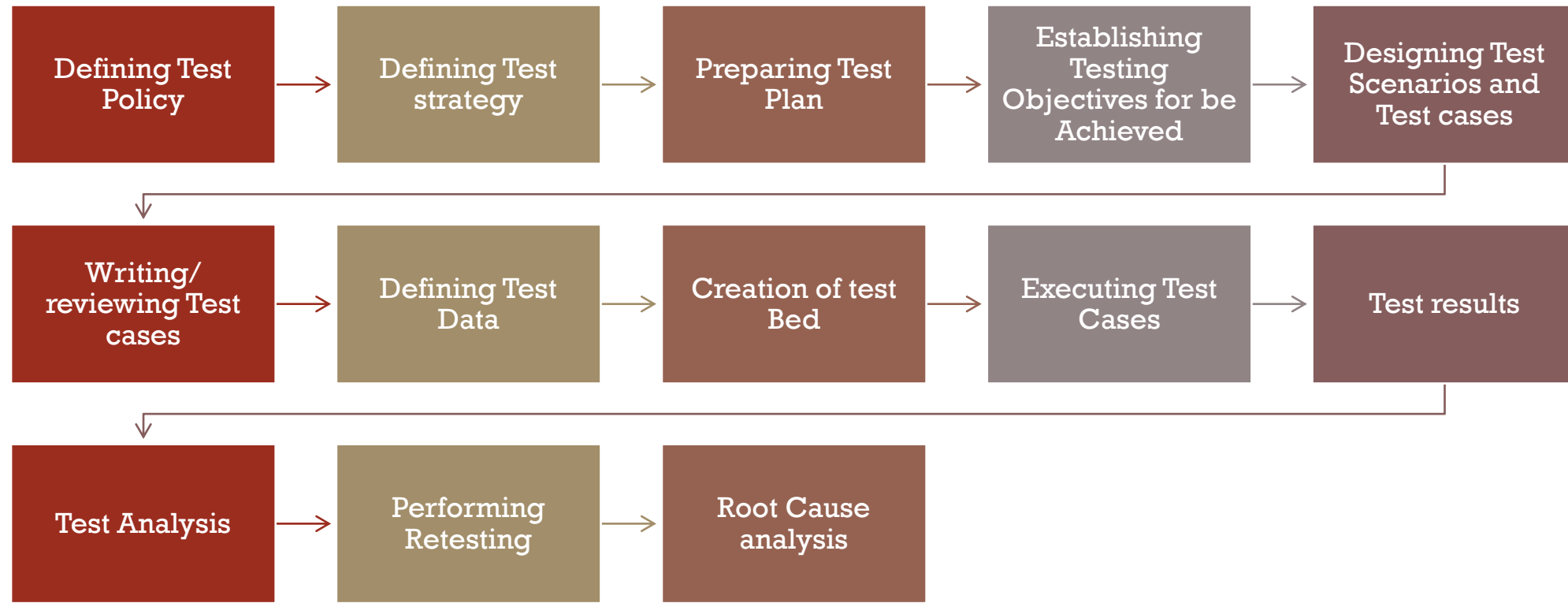
DEVELOPING TEST STRATEGY

- A test strategy document is a high level document, developed by project manager
- It is normally derived from the business requirement specification Document
- It is not updated too often(Static Document)
- Process:
 - Select & Rank test factors for given application
 - Identify system development phases and related test factors
 - Identify associated risks with each selected test factor in case if it is not achieved
 - Identify phases in which risks of not meeting a test factor need to be addressed

DEVELOPING TESTING METHODOLOGIES(PLAN)

- Acquire and study test strategy
- Determine the type of development project being executed
- Determine the type of software system being made
- Identify tactical risks related to development
- Determine when testing must occur during life cycle
- Steps to develop customised test strategy
- Type of development methodology impact test plan decisions

TESTING PROCESS



ATTITUDE TOWARDS TESTING

- It starts from test policy and ends to test strategy definition
- Cross-cultural Differences
- Lack of proper communication
- Developers may not accept the defects in first place
- Conflict between developer and tester

RAISING MANAGEMENT AWARENESS FOR TESTING

Management should be made aware of the importance of testing a product because the funds required will be provided by them.

Without adequate funding for testing, software quality suffers.

Benefits of S/W testing are enormous

- Quality
- Satisfied Customer
- Bringing Profit
- User Experience
- Business Optimisation

SKILLS REQUIRED BY TESTER

- Understanding Business needs
- Good communication skills
- Concepts of Testing
- Levels of Testing
- Knowing what to prioritize
- Continuous Learning
- Techniques for validation and verification
- Risk assessment and management
- Developing Test plan
- Execution of test plan

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